

Suryansh Singh Rawat

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EDUCATION

University of Washington, Seattle

Master of Science (M.S), Data Science | **GPA: 3.81/4.0**

Sept 2025 – Mar 2027

Seattle, WA

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering (B.E), Electronics and Instrumentation

Aug 2018 – July 2022

Goa, India

TECHNICAL SKILLS

Languages: Python, C++, SQL, R, Java, TypeScript, Bash, MATLAB

Developer Tools: Git, AWS, GCP, Azure, Docker, S3, DynamoDB, PostgreSQL, MongoDB, Redis, Supabase

Frameworks: PyTorch, TensorFlow, Keras, SpaCy, HuggingFace, LangChain, LangGraph, Streamlit, FastAPI, Selenium

WORK EXPERIENCE

Ascentt

Mar 2024 – Sept 2025

Principal AI Engineer

Remote, US

- Built and scaled a ChatGPT-like enterprise AI platform for a **Fortune 500 automotive company** from **0 to 56K+** users; developed core **REST APIs** and **DynamoDB**-backed user state infrastructure.
- Architected **Agentic RAG** pipeline using **LangGraph** managing **1TB+ of unstructured data** across **14+** document types.
- Optimized RAG pipelines across **4 enterprise** products, implementing lossless preprocessing, chunking, and indexing in **Amazon OpenSearch**, and benchmarking with **RAGAS**, achieving **+25% Precision@10** and **+12% Recall@10** over vanilla RAG baseline.
- Owned secure file ingestion for enterprise AI systems, designing **encrypted upload** and preprocessing pipelines using **AES-GCM with AWS KMS**, enabling compliant handling of sensitive documents across multiple file types.
- Led backend repository maintenance** and code reviews, ensuring high-quality, production-ready code across enterprise systems.

Deloitte USI

July 2022 – Mar 2024

Data Scientist | AI Center of Excellence Team

Bangalore, India

- Developed **Autoencoder anomaly detection** models with graph-based preprocessing in **Neo4J** to detect **zero-day threat attacks** from real-time AWS network flow data.
- Researched an NLP pipeline for regulatory document consolidation, leveraging **metric learning** trained **SecBERT** embeddings and **HDBSCAN** to merge semantically overlapping compliance requirements, achieving **~83% reduction** in manual SME efforts.
- Constructed a **SecBERT text summarization model** for generating cyber threat reports, reducing analyst review time by **~40%**, integrating **NER tagging and masking** with **SpaCy** annotation, deploying end-to-end pipeline using **Flask and Docker**.
- Built **15+ web scrapers** using **Selenium** and **BeautifulSoup** to collect large-scale cybersecurity threat reports for model training.

Scienaptic AI

July 2021 – Dec 2021

Data Scientist Intern

Bangalore, India

- Achieved an **11.5% increase** in application approvals by building credit underwriting models using **XGBoost and Logistic Regression** to forecast the probability of credit accounts that may default.

SELECTED PROJECTS

Ad-Astra: xAI Ad Intelligence Engine 🧠 | Python, Grok, XAPI, Supabase

Dec 2025 – Dec 2025

- Built a personalized ad generation and placement engine for X (Twitter), using **Grok 4.1 Reasoning & Vision models** for persona-aware ad copy generation and **scroll telemetry** to optimize in-feed placement; improving engagement and ad relevance.

OpenProbe 🧠 | Python, Langchain, LangGraph

May 2025 – June 2025

- Built OpenProbe, a deep research **multi agent system** using **LangGraph** for state orchestration (websearch, code, reasoning).
- Achieved **67.1% accuracy** on the FRAMES benchmark, outperforming GPT-4o-Search by **1.5%**, using DeepSeek-R1/Qwen3-32B.

PersonaAI 🧠 | Python, Microsoft Phi-4, Supabase

Oct 2025 – Oct 2025

- Built an **AI simulation platform** to test digital campaigns before launch, running **A/B tests** using **NVIDIA Nemotron** Personas.
- Created natural-language audience targeting with diagnostics explaining why variants succeed or fail across segments.

KeyCognition 🧠 | Python, PyTorch, React

Nov 2025 – Nov 2025

- Built a behavioural analysis platform processing **136M+ keystroke events** to predict user cognitive load (low/med/high).
- Leveraged temporal typing features (e.g. using backspace) with **LSTM and K-Means models** for user behavior classification.

ACHIEVEMENTS

- Placed **3rd out of 40+ teams** at the **Lovable.dev Seattle Hackathon 2025** for building **PersonaAI**.
- Placed **3rd out of 20+ teams** at the **UW Databricks Hackathon 2025** for building **KeyCognition**.
- Recipient of **\$5000 Thinking Machines Research Grant** for proposed research on improving LLM Planning and Personalization.